ie UNIVERSITY

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

MCSBT

Master in Computer Science & Business Technology

john E. Ortega, PhD | Fall 2026 Semester 1 – Week 5 Session 11



Class Plan

Session #	Date	Topic
1	**Sep 28**	AI Foundations and Technology Landscape
2	**Sep 30**	AI Strategy and Business Transformation
3	**Oct 2**	Machine Learning Fundamentals (assignment 1)
4	**Oct 5**	Deep Learning
5	**Oct 7**	NLP and Foundation Models (assignment 2)
6	**Oct 14**	Large Language Models (Quiz 1)
7	**Oct 16**	Retrieval-Augmented Generation (RAG)
8	**Oct 19**	Agentic AI Systems (assignment 3)
9	**Oct 21**	Multimodal AI (Midterm)
10	**Oct 26**	Recommender Systems (Quiz 2,Final Projects)
11	**Oct 28**	Cloud AI, MLOps and LLMOps (assignment 4, group sel)
12	**Oct 30**	Responsible AI and Governance
13	**Nov 4**	Emerging Topics (assignment 5)
14-15	**Nov 11 + 13**	Final Presentations

Session 11 Agenda

- Deployment
- Monitoring
- Observability
- Evaluation
- Cost Optimization
- AI Security
- Azure AI Foundry, AWS Bedrock, Vertex AI, Databricks, Snowflake AI
- Assignment 4
- Group Selection



Session 11

Cloud AI, MLOps and LLMOps

Deployment

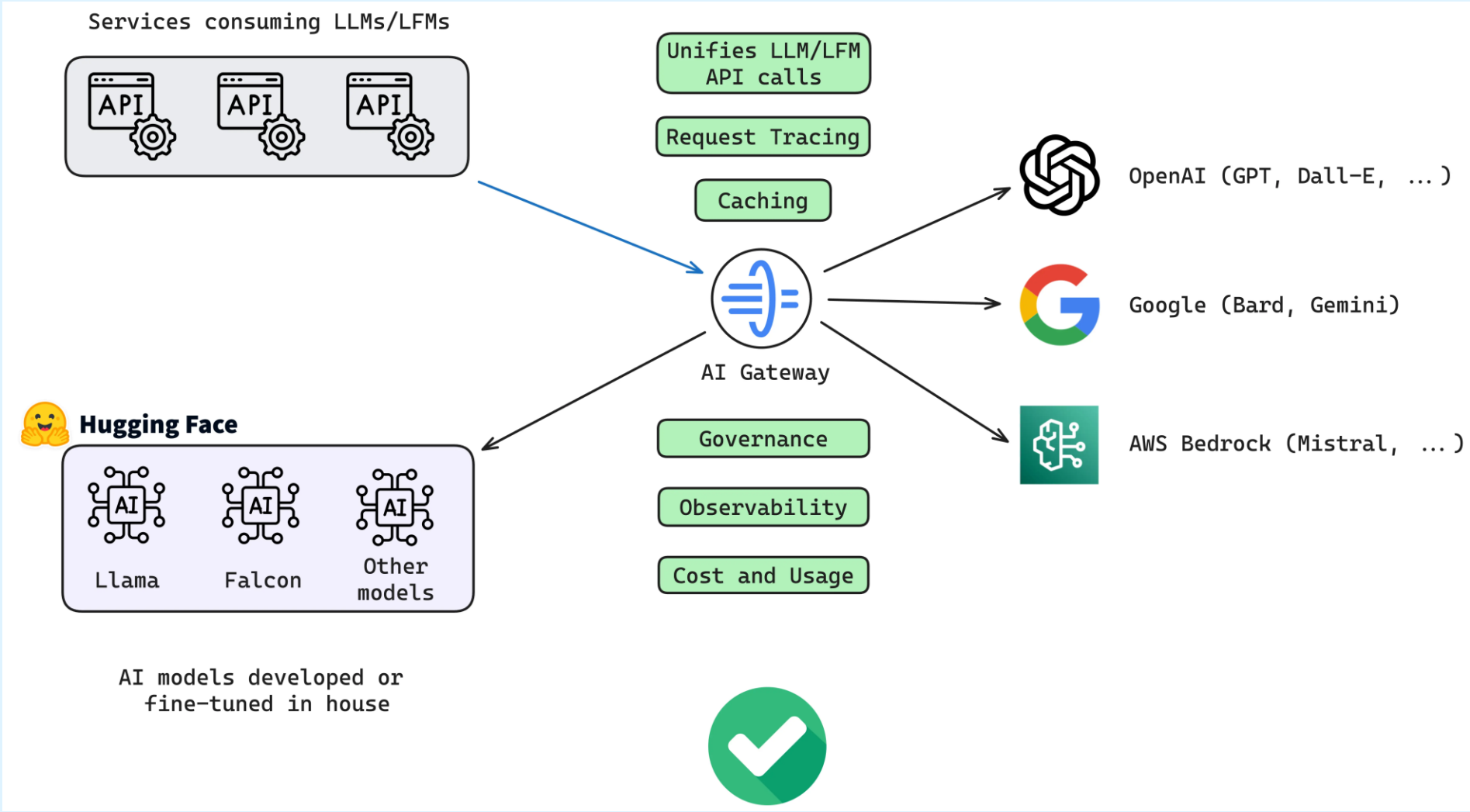
Deployment: What Is It?

Deployment turns a trained model into a reliable service other software can call.

- **Package:** model weights, code, and dependencies ship together in a container
- **Serve:** an inference server exposes the model as an API endpoint
- **Release safely:** canary or blue-green rollouts and autoscaling limit the blast radius

Source: Huyen, Designing Machine Learning Systems (2022).

Deploying LLM Services Into the Cloud



Deployment: Production Architecture



- **Gateway:** handles authentication, rate limits, and routing between model versions
- **Servers:** continuous batching on GPU pods keeps throughput high; autoscaling follows queue depth
- **Registry:** versioned artifacts allow canary releases and instant rollback

Source: KServe and vLLM documentation; typical Kubernetes serving pattern.

Deployment: API Code Example

Serve an open model behind an OpenAI-compatible API with vLLM, then call it.

```
# shell
vllm serve meta-llama/Llama-3.1-8B-Instruct --max-model-len 8192

# python client
from openai import OpenAI
client = OpenAI(base_url="http://localhost:8000/v1", api_key="none")
r = client.chat.completions.create(
    model="meta-llama/Llama-3.1-8B-Instruct",
    messages=[{"role": "user", "content": "Hello!"}])
print(r.choices[0].message.content)
```

Source: vLLM documentation.

Deployment: Tools

Managed commercial services on the left, open-source projects on the right.

Commercial	Open source
Amazon SageMaker endpoints	vLLM
Azure ML managed online endpoints	KServe
Google Vertex AI endpoints	BentoML
Databricks Model Serving	Ray Serve
NVIDIA NIM / Triton	Hugging Face TGI



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Cloud AI, MLOps and LLMOps

Monitoring

Monitoring: What Is It?

Monitoring tracks the health of a live model so problems are caught before users report them.

- **System metrics:** latency, throughput, error rate, and GPU utilization
- **Model metrics:** input drift, prediction drift, and quality scores over time
- **Alerts:** thresholds page the on-call engineer when a metric degrades

Source: Sculley et al., Hidden Technical Debt in ML Systems (2015).

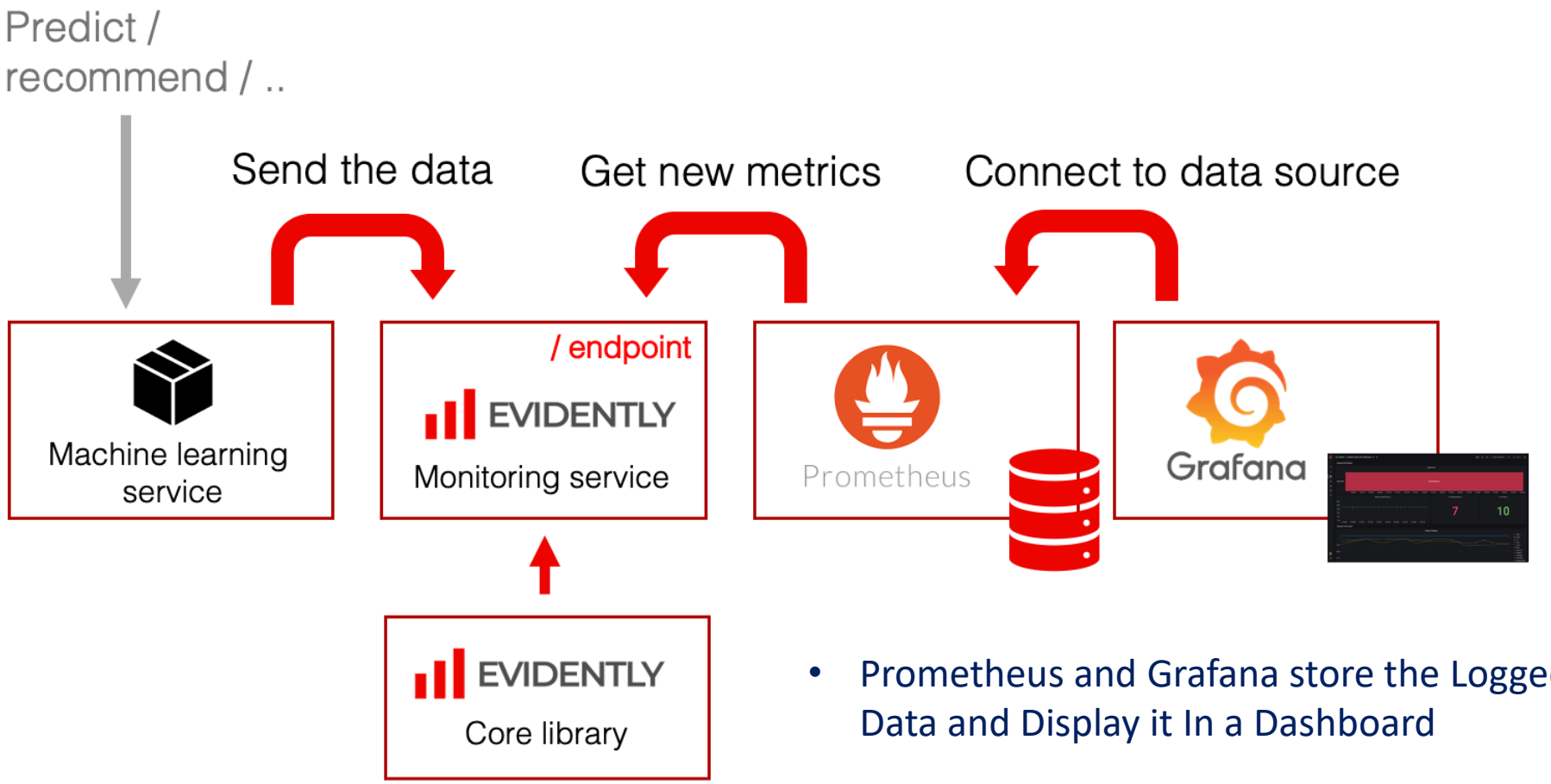
Monitoring: Production Architecture



- **Collect:** services expose counters and histograms that a scraper pulls every few seconds
- **Visualize:** dashboards show p50/p95 latency, token usage, and drift side by side
- **Respond:** alerts route to Slack or PagerDuty with a runbook link

Source: Prometheus and Grafana documentation.

Monitoring System Design



- Prometheus and Grafana store the Logged Data and Display it In a Dashboard

Monitoring: Code Example

Expose latency and request counts from a model endpoint with Prometheus.

```
from prometheus_client import Counter, Histogram, start_http_server

REQS = Counter("llm_requests_total", "Requests", ["status"])
LAT = Histogram("llm_latency_seconds", "Latency")

start_http_server(9100)

@LAT.time()
def handle(prompt):
    ...  # call the model; REQS.labels("ok").inc()
```

Source: prometheus_client documentation.

Monitoring: Tools

Managed commercial services on the left, open-source projects on the right.

Commercial	Open source
Datadog	Prometheus
New Relic	Grafana
Arize AI	Evidently AI
Fiddler AI	whylogs / WhyLabs



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Cloud AI, MLOps and LLMOps

Observability

Observability: What Is It?

Observability lets you ask why an LLM application behaved a certain way, not just whether it is up.

- **Traces:** each request is recorded as spans: retrieval, prompt, model call, tool call
- **Context:** prompts, outputs, token counts, and cost are stored with the trace
- **Debugging:** replay a bad answer end to end to find the failing step

Source: OpenTelemetry GenAI semantic conventions.

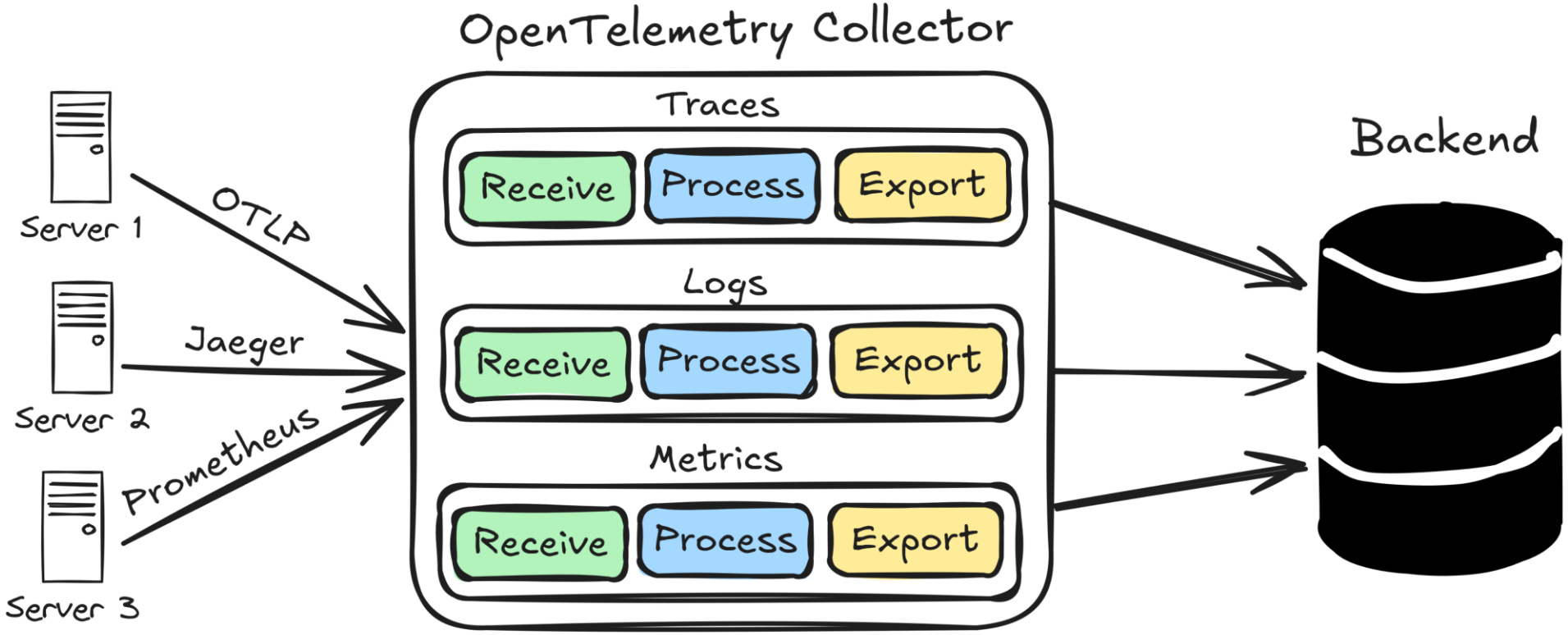
Observability: Production Architecture



- **Instrument:** an SDK wraps functions and model calls and emits spans
- **Standardize:** OpenTelemetry keeps traces portable across vendors
- **Close the loop:** user feedback and failing traces become new evaluation cases

Source: OpenTelemetry and Langfuse documentation.

Observability: Open Telemetry Collector



Observability: Code Example

Trace a RAG function with the Langfuse decorator.

```
from langfuse import observe

@observe()
def retrieve(q):
    return vector_db.search(q, k=4)

@observe()
def answer(q):
    docs = retrieve(q)
    return llm(q, context=docs)
```

Source: Langfuse documentation.

Observability: Tools

Managed commercial services on the left, open-source projects on the right.

Commercial	Open source
LangSmith	Langfuse
Datadog LLM Observability	Arize Phoenix
Weights & Biases Weave	OpenTelemetry
Braintrust	MLflow Tracing

Source: vendor and project documentation.



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Cloud AI, MLOps and LLMOps

Evaluation

Evaluation: What Is It?

Evaluation measures answer quality with repeatable tests, before release and while in production.

- **Offline:** a golden dataset of questions and reference answers runs on every change
- **LLM-as-judge:** a strong model scores relevance, faithfulness, and tone
- **Online:** user feedback and sampled production traces catch what tests missed

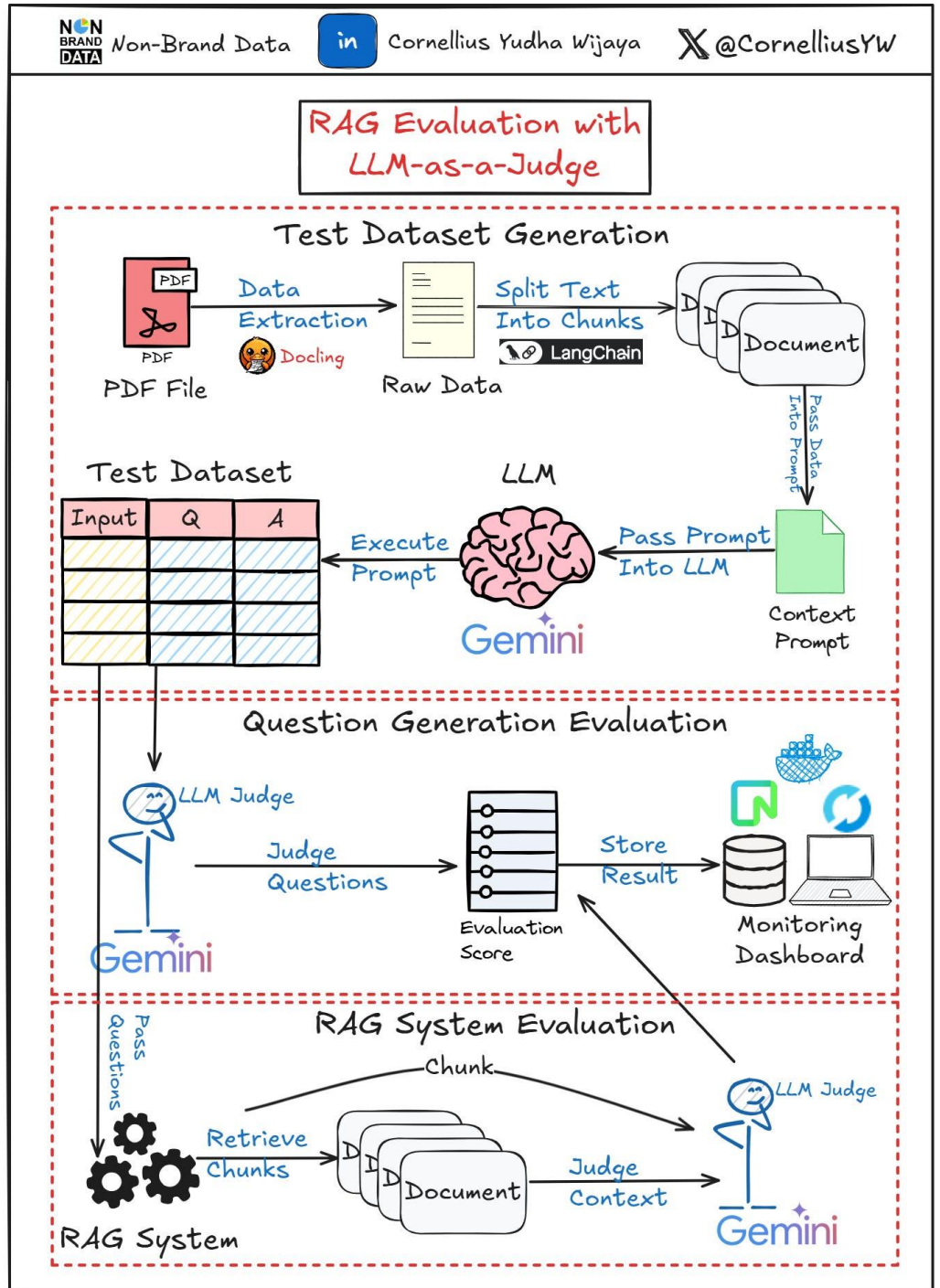
Source: Zheng et al., Judging LLM-as-a-Judge (2023).

Evaluation: Production Architecture



- **Gate releases:** a pull request fails if relevance or faithfulness drops below a threshold
- **Version everything:** prompt, model, and dataset versions make results comparable
- **Watch judge bias:** spot-check judge scores against human labels

Source: DeepEval and Ragas documentation.



Evaluation: LLM-as-a-JUDGE

- Example RAG Evaluation using LLM-as-a-Judge
- Set of questions are created with Answers
- LLM is used with the test data set and questions answers
- Results are stored and scored

Evaluation: Code Example

Score an answer for relevance with DeepEval, usable as a pytest check.

```
from deepeval.metrics import AnswerRelevancyMetric
from deepeval.test_case import LLMTestCase

case = LLMTestCase(
    input="What is the refund window?",
    actual_output="Refunds are accepted within 30 days.")

metric = AnswerRelevancyMetric(threshold=0.7)
metric.measure(case)
print(metric.score, metric.is_successful())
```

Source: DeepEval documentation.

Evaluation: Tools

Managed commercial services on the left, open-source projects on the right.

Commercial	Open source
LangSmith	DeepEval
Braintrust	Ragas
Arize AI	promptfoo
Patronus AI	OpenAI Evals / lm-eval-harness

Source: vendor and project documentation.



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Cloud AI, MLOps and LLMOps

Cost Optimization

Cost Optimization: What Is It?

LLM bills grow with tokens and model size, so cost is managed like any other engineering budget.

- **Cost drivers:** input tokens, output tokens, model tier, and call volume
- **Levers:** caching, routing easy queries to small models, batching, shorter prompts
- **Self-hosting:** quantized open models cut unit cost at high, steady volume

Source: Chen et al., FrugalGPT (2023).

Cost Optimization: Production Architecture



- **Cache first:** a cache hit costs nothing and answers in milliseconds
- **Route:** a cheap classifier sends simple questions to a small model
- **Track:** per-feature and per-user token budgets expose runaway spend

Source: FrugalGPT; LiteLLM and GPTCache documentation.

Cost Optimization: Code Example

A minimal cache plus router in plain Python.

```
cache = {}

def ask(q):
    if q in cache:
        return cache[q]
    model = "gpt-4.1-mini" if len(q) < 200 else "gpt-4.1"
    out = llm(model, q)
    cache[q] = out
    return out
```

Source: illustrative pattern; see LiteLLM Router for production use.

Cost Optimization: Tools

Managed commercial services on the left, open-source projects on the right.

Commercial	Open source
OpenAI / Anthropic Batch APIs	LiteLLM
AWS Bedrock Intelligent Prompt Routing	GPTCache
Portkey	Helicone
OpenRouter	vLLM / llama.cpp

Source: vendor and project documentation.



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Cloud AI, MLOps and LLMOps

AI Security

AI Security: What It Is

AI security protects models, data, and tools from attack and from unintended behavior.

- **Prompt injection:** hostile text in a prompt or document overrides instructions
- **Data leakage:** secrets or personal data appear in prompts, outputs, or logs
- **Excess agency:** agents with broad tool permissions can cause real damage

Source: OWASP Top 10 for LLM Applications (2025).

AI Security: Production Architecture



- **Filter both ways:** screen inputs for injection and outputs for PII or unsafe content
- **Limit blast radius:** give tools the minimum permissions and require approval for risky actions
- **Audit:** log every prompt, tool call, and policy decision; red-team regularly

Source: OWASP LLM Top 10; NIST AI Risk Management Framework.

AI Security: Code Example

Mask personal data before text reaches a model, using Microsoft Presidio.

```
from presidio_analyzer import AnalyzerEngine
from presidio_anonymizer import AnonymizerEngine

text = "I am Jane Roe, call me at 212-555-0147."
found = AnalyzerEngine().analyze(text=text, language="en")
safe = AnonymizerEngine().anonymize(
    text=text, analyzer_results=found)
print(safe.text)  # I am <PERSON>, call me at <PHONE_NUMBER>.
```

Source: Microsoft Presidio documentation.

AI Security: Tools

Managed commercial services on the left, open-source projects on the right.

Commercial	Open source
Lakera Guard	NeMo Guardrails
AWS Bedrock Guardrails	Llama Guard
Azure AI Content Safety	Guardrails AI
Protect AI	Presidio / garak

Source: vendor and project documentation.



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Cloud AI, MLOps and LLMOps

Azure AI Foundry, AWS Bedrock, Vertex AI, Databricks, Snowflake AI

Cloud AI Platforms: What It Is

Cloud AI platforms bundle model access, customization, deployment, and governance in one managed service.

- **Model catalog:** hosted frontier and open models behind a single API
- **Build:** fine-tuning, RAG, agents, and evaluation tooling
- **Govern:** identity, private networking, audit logs, and data residency

Source: Azure, AWS, Google Cloud, Databricks, and Snowflake documentation.

Cloud AI Platforms: Production Architecture



- **Data gravity:** platforms differ mostly in where the data already lives
- **Same pattern:** every vendor offers catalog, endpoint, guardrails, and evaluation
- **Trade-off:** convenience and compliance versus lock-in and per-token pricing

Source: generic managed-platform reference architecture.

Cloud AI Platforms: Code Example

Call a hosted model on AWS Bedrock; the other platforms follow the same shape.

```
import boto3

br = boto3.client("bedrock-runtime", region_name="us-east-1")
resp = br.converse(
    modelId="anthropic.claude-3-haiku-20240307-v1:0",
    messages=[{"role": "user", "content":
              [{"text": "Summarize our refund policy."}]}],
    inferenceConfig={"maxTokens": 300})
print(resp["output"]["message"]["content"][0]["text"])
```

Source: AWS Bedrock Converse API documentation.

Cloud AI Platforms: Platform Comparison

Five managed platforms, each strongest where your data already lives.

Platform	Provider	Strength
Azure AI Foundry	Microsoft	OpenAI models, Entra ID
AWS Bedrock	Amazon	Many vendors, one API
Vertex AI	Google	Gemini, BigQuery
Databricks Mosaic AI	Databricks	Lakehouse, Unity Catalog
Snowflake Cortex AI	Snowflake	AI inside the warehouse



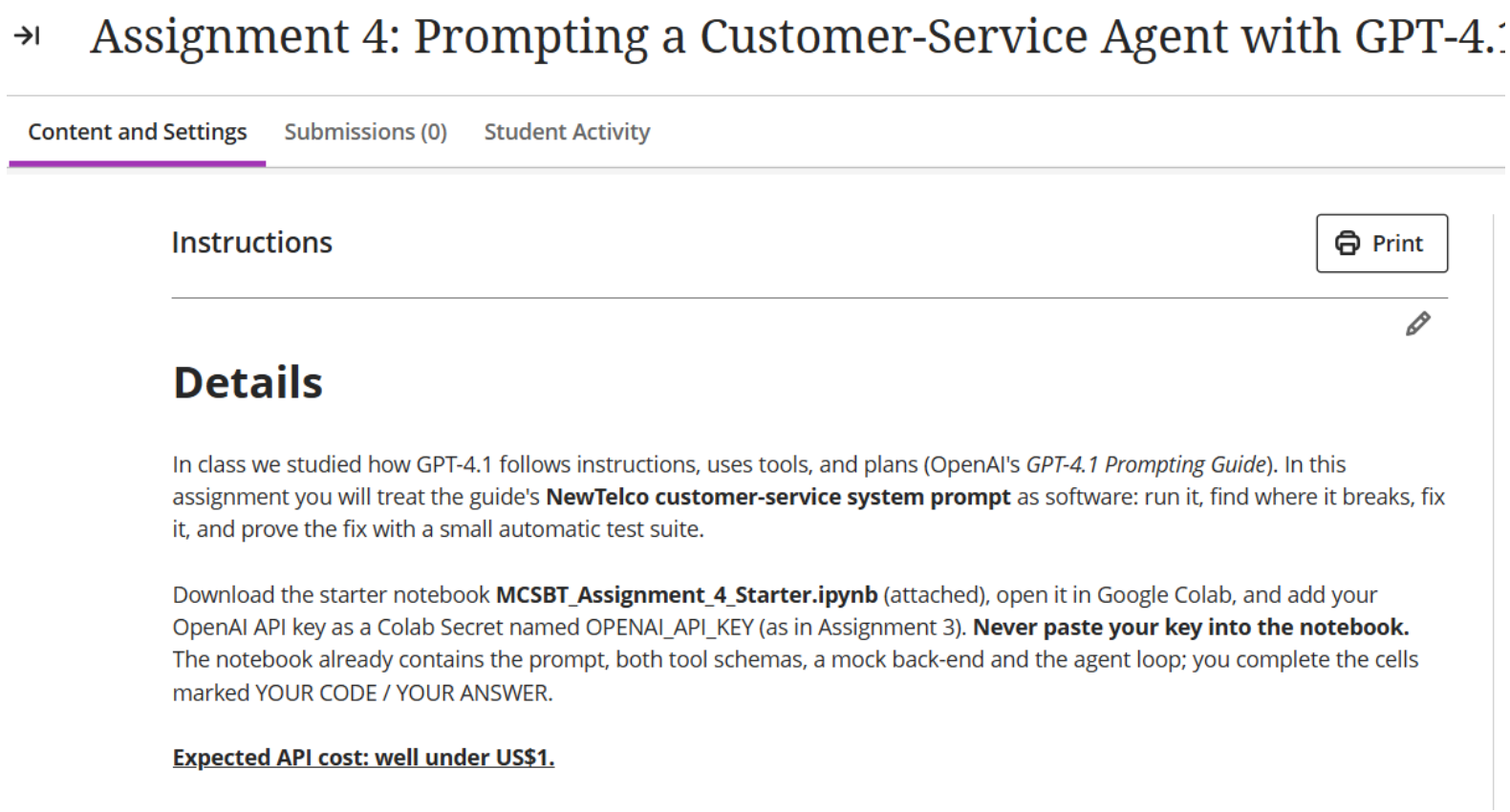
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Cloud AI, MLOps and LLMOps

Assignment 4

Assignment 4 Available Online

- Due Friday, November 6th.
- Requires OpenAI Key
- Cost is around $1



Assignment 4: Prompting a Customer-Service Agent with GPT-4.1

Content and Settings | Submissions (0) | Student Activity

Instructions

Print

Details

In class we studied how GPT-4.1 follows instructions, uses tools, and plans (OpenAI's *GPT-4.1 Prompting Guide*). In this assignment you will treat the guide's **NewTelco customer-service system prompt** as software: run it, find where it breaks, fix it, and prove the fix with a small automatic test suite.

Download the starter notebook **MCSBT_Assignment_4_Starter.ipynb** (attached), open it in Google Colab, and add your OpenAI API key as a Colab Secret named OPENAI_API_KEY (as in Assignment 3). **Never paste your key into the notebook.** The notebook already contains the prompt, both tool schemas, a mock back-end and the agent loop; you complete the cells marked YOUR CODE / YOUR ANSWER.

Expected API cost: well under US$1.



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Cloud AI, MLOps and LLMOps

Group Selections

Group Selections

- **PLEASE REPLY ON THE DISCUSSION BOARD WITH YOUR ANTICIPATED GROUP**
- **IF YOU DO NOT HAVE A GROUP BY FRIDAY, GROUPS WILL BE ASSIGNED**

ZAID ABDELQADER
LATIKO MULUGETA AGA
ALEJANDRO CAMPA MARTÍNEZ
VINSENSIUS FERDINANDO
PATRICIA GABRIELLE GARCIA
MARC GABRIEL HABIB

DINGSHUANG HOU
JULIO ALBERTO LLORT TEJEDA
AKSHAY MAHESH IYER
MATTEO PIO MELONE
SOHITH MINDI
ANDRÉS ALEJANDRO MIRAZO BALLESTEROS

ISSAGO MOSIKARE
DANIYAL NAINITALWALA JUNAID
LUCÍA PARREÑO LEGORBURO
JORGE PLAZA DE LOS REYES ÁLVAREZ
EMMA ROYO BOATELLA
KARIM SAHILI HELBAWI
JACK SANGSTER
KARIMA TAHA EL BABA
KONARK VINOD
VIDAK VUKADINOVIC
OSCAR WEISHUT
YOHANNES BELAY WOLDEYES

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Thank you

Questions & discussion

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